

DATA ANALYSIS INTERNSHIP

Day 2 – REPORT

On Day 2, the concept of **Exploratory Data Analysis (EDA)** was discussed and performed using a **Student Performance Dataset** to analyze factors affecting student results such as study hours, attendance, sleep, and exam scores.

1. Dataset Collection

The **Student Performance Dataset** was used for performing **Exploratory Data Analysis (EDA)**.

The dataset contained information related to student academic performance and behavioral factors. The required columns were selected for analysis.

The final dataset contains:

- Student ID
 - Hours Studied
 - Sleep Hours
 - Attendance Percentage
 - Previous Scores
 - Exam Score
 - Result (Pass/Fail)
-

2. Dataset Columns

STUDENT PERFORMANCE ANALYSIS (EXPLORATORY DATA ANALYSIS)

Column	Contains
student_id	Unique student identification values (String)
hours_studied	Number of study hours such as 1–12 hours (Float)
sleep_hours	Student sleep duration in hours (Float)
attendance_percent	Attendance percentage values like 60%, 75%, 90% (Float)

previous_scores	Previous academic scores (Integer)
exam_score	Final examination marks (Float)
result	Performance category such as Pass, Fail (String)

3. Exploratory Data Analysis (EDA)

Performed **EDA** to understand student performance patterns and relationships between study behavior and exam results.

EDA Steps Performed

- Checked dataset shape
 - Identified rows and columns
 - Verified data types
 - Checked missing values
 - Checked duplicate values
 - Performed summary statistics analysis
 - Analyzed target variable distribution
 - Studied correlation between features
 - Compared student study habits with results
-

4. Dataset Overview

The dataset contains a sample of **50 student records**.

Features Included:

- Hours Studied
- Sleep Hours
- Attendance Percentage
- Previous Scores
- Exam Score
- Result (Pass/Fail)

Target Variable: Result

The **Result** column contains:

- **Pass – 23 students**
- **Fail – 27 students**

Observation:

The dataset is relatively balanced, but slightly skewed toward the **Fail** category.

5. Summary Statistics

Average Values

Feature	Average Value
Hours Studied	~5.96 hours
Sleep Hours	~6.62 hours
Attendance Percentage	~75.65%
Previous Scores	~68.18
Exam Score	~33.54

Observation:

- Students generally study **moderate hours (~6 hours average)**.
 - Exam scores are relatively low, with an average score of around **33 marks**.
-

6. Distribution Analysis

Hours Studied

- Study hours range between **1 to 12 hours**.
- Most students study between **3–10 hours**.
- Students with **higher study hours** are more likely to **pass**.

Exam Score

- Most exam scores lie between **30–40 marks**.
- Very few students have extremely low or very high scores.
- Score distribution shows a slight tendency toward higher values.

Observation:

Higher study time positively influences examination performance.

7. Correlation Analysis

Strong Relationships

Hours Studied → Exam Score

Correlation: 0.79 (Very Strong Positive Correlation)

Observation: More study hours result in higher exam scores.

Exam Score → Result

Correlation: 0.77 (Strong Correlation)

Observation: Exam score strongly determines whether a student passes or fails.

Hours Studied → Result

Correlation: 0.67 (Strong Positive Correlation)

Observation: Study hours have a direct impact on academic performance.

Moderate Relationships

- Previous Scores → Exam Score (0.36)
- Previous Scores → Result (0.32)

Weak Relationships

- Sleep Hours → Very weak impact on result
- Attendance Percentage → Weak impact (~0.15)

8. Data Types

Column	Data Type
student_id	String
hours_studied	Float
sleep_hours	Float
attendance_percent	Float
previous_scores	Integer
exam_score	Float
result	String

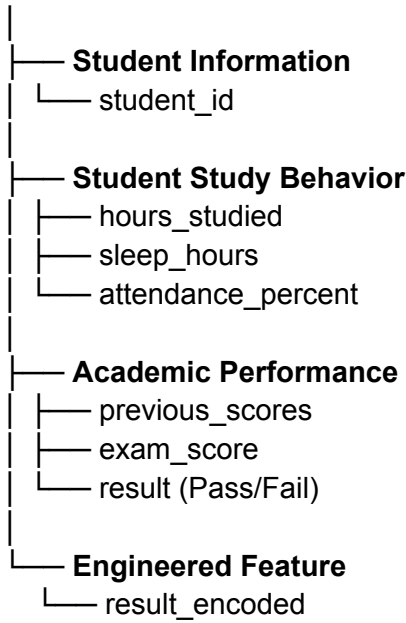
9. Feature Role Analysis

Column	Role
student_id	Identifier
hours_studied	Feature
sleep_hours	Feature
attendance_perce nt	Feature
previous_scores	Feature
exam_score	Target
result	Target
result_encoded	Engineered Target

10. Tree Structure Creation

The tree structure was created to show the organization of the **Student Performance Dataset** in a hierarchical format.

Student Performance Dataset



Observation:

The **exam score** has the strongest impact on determining student results, while **hours studied** is the most important predictor of success.

11. Key Business / Machine Learning Insights

Most Important Feature

Hours Studied is the strongest predictor of student success.

Less Important Features

- Attendance Percentage showed only a weak impact.
- Sleep Hours had minimal influence in this dataset.

Model Implication

If building a classification model, the best features would be:

- **hours_studied**
 - **exam_score**
 - **previous_scores**
-

12. Conclusion

On Day 2, I learned the concept of **Exploratory Data Analysis (EDA)** and performed EDA on a **Student Performance Dataset**. I analyzed important factors such as **study hours, sleep hours, attendance percentage, previous scores, and exam scores** to understand their impact on student results. I also studied **summary statistics, data distribution, correlations, data types, and feature importance**. The analysis showed that **hours studied and exam scores strongly influence student performance and final results**.